Team Details

Team name: Team Zen

Team leader name: Hiran Vikraman S R

Team size: 2   (Hiran Vikraman S R · Arun G K)

Problem Statement: Challenge 1. Intelligent Conversational AI for the KSP Crime Database. Enable officers to query the FIR/crime database in natural language (English and Kannada) and get accurate, evidence-grounded answers.

Brief about the solution

ANVESHAK (ಅನ್ವೇಷಕ = "the investigator") turns the KSP crime database into an **AI investigation partner**, not just a chatbot.

• Ask in Kannada or English, by voice or text, and get an answer in seconds **with the evidence attached**: the SQL, the rows, the case IDs. No SQL skills, no analyst queue.

• **Deterministic tools compute; the model only narrates.** Every number traces to a tool result, so there are no invented statistics. That is what makes it usable in policing.

• It does not stop at answers, it works cases: links serial crime across district boundaries, files new FIRs and links them live, runs a six-agent cell that assembles court-ready packs, and sweeps overnight to raise leads before anyone asks.

• **Impact: minutes instead of days, catches that siloed records miss, and a ranked lead list every morning. Runs 100% on Zoho Catalyst.**

Opportunities

How it is different, and how it solves the problem:

• Most "crime chatbots" stop at natural language to SQL. ANVESHAK adds an investigation layer: cross-district serial linkage, live FIR intake, multi-agent case work and proactive patrol.

• **Trust by construction:** the model never invents numbers, the linkage ground truth is public, role scope is enforced in the query, and the audit log is tamper-evident. There is a red-team console in the product so anyone can test it.

• **Cross-district reach:** MO fingerprinting via multilingual embeddings finds serial offenders that siloed district records miss.

• **Built for real officers:** bilingual Kannada and English, voice or text, government-grade UI.

Where it makes a difference: a constable at a rural station gets a statewide answer in Kannada in seconds; an SP spots a ring crossing Mandya into Bengaluru before the next strike; an IO gets a first-draft court-ready pack in minutes; a control room sees tomorrow's hotspot tonight.

USP: "Ask a question, get a verified answer with evidence, and let the AI work the case end to end to a court-ready pack."

List of features offered by the solution

1. **Conversational floor**: bilingual English and Kannada, voice or text, verified natural-language-to-SQL with an evidence drawer on every answer.
2. **Serial Crime Linkage Engine**: MO fingerprinting (multilingual embeddings plus structured features), weighted cosine, HDBSCAN, 180-day / 120-km filter. Discovers cross-district series cold.
3. **Live FIR intake**: file a new FIR in plain words, typed or dictated in Kannada. It is embedded at runtime and joins a matching series immediately.
4. **AI Investigation Cell**: six agents stream their reasoning and assemble a court-ready Investigation Pack, rendered to PDF by SmartBrowz.
5. **CrimeGraph and Person 360**: a knowledge graph over people and cases, plus one page holding a person's whole footprint, risk breakdown and associates.
6. **Night Patrol and patrol plans**: spike, series-growth and repeat-offender detectors produce ranked lead cards, then a deployment plan per district.
7. **Trust Center**: live metrics, a red-team console anyone can attack, and a hash-chained audit log that proves history has not been rewritten.
8. **Governance**: role scope enforced server-side (SHO, SP, SCRB, analyst with masked names), and no protected attributes in any model (ADR-9).

Process flow diagram or Use-case diagram

Officer (English or Kannada, voice or text) to React SPA to a FastAPI intent router:

- **Question path:** schema card and few-shots, then guardrails (SELECT-only, allowlisted tables, no file IO), the ADR-9 policy screen, role-scope injection, and self-repair. Answer plus evidence drawer.
- **Tool path:** linkage, graph, forecast, hotspots, risk and similarity tools compute on DuckDB; the model narrates the verified result.
- **New FIR:** the narrative is embedded at runtime by ONNX, written to Data Store and the mirror, linkage re-runs, and the officer is told which series it joined.
- **Investigation:** "Investigate SH-07" starts the six-agent pipeline over SSE, ending in the Investigation Pack and its PDF.
- **Autonomous:** Cron runs Night Patrol, which produces lead cards, a patrol plan and a spoken Kannada morning briefing.

Two data layers (ADR-1): Catalyst Data Store is the system of record; an embedded DuckDB mirror serves sub-second analytics.

Wireframes/Mock diagrams of the proposed solution (optional)

Nine views in the live prototype (government-grade dark theme):

- **Chat**: bilingual answer, evidence drawer, voice input, scope badge.
- **Lead Feed**: ranked Night-Patrol cards, patrol plan, spoken briefing.
- **Series**: discovered series with codename, link explanations, replay and the counterfactual banner.
- **New FIR**: file a case in plain words and watch it join a series.
- **Investigation Room**: six agents streaming, then the assembled pack.
- **Person 360**: one person's history, risk breakdown, network, associates.
- **CrimeGraph**: interactive people and case network.
- **Trust Center**: metrics, red-team console, audit-chain verification.
- **Audit**: every action logged with its role.

Live: https://anveshak-api-50044329134.development.catalystappsail.in/ui/

Architecture diagram of the proposed solution

Officer (English / Kannada, voice and text)

React SPA, served by the backend at /ui

FastAPI on Catalyst AppSail, then the intent router

guardrails, ADR-9 policy, role scope, self-repair, and typed tools

(linkage, graph, forecast, hotspots, risk, similarity)

DuckDB analytical mirror, with Catalyst Data Store as system of record

ONNX MiniLM embeds new text at runtime, with no external model calls

LLM adapter to Catalyst QuickML, serving GLM-4.7-Flash

Six-agent Investigation Cell over SSE, then SmartBrowz for the pack PDF

Cron drives Night Patrol, producing lead cards and the morning briefing

Deterministic tools compute, the model narrates (ADR-2). No external inference API is ever called from the deployed app (ADR-4).

Technologies to be used in the solution

Backend: Python 3.12, FastAPI, DuckDB, sqlglot, ONNX Runtime with paraphrase-multilingual-MiniLM-L12-v2, scikit-learn and HDBSCAN, NetworkX with Louvain, statsmodels SARIMA, sse-starlette.

Frontend: React 18, Vite, Tailwind CSS, ECharts for charts and the network graph, Leaflet with OpenStreetMap, Web Speech API for kn-IN voice.

LLM: GLM-4.7-Flash served through Catalyst QuickML LLM Serving.

Packaging: Docker custom AppSail runtime, with the scientific stack, the embedding model and the SPA bundled in the image.

Quality: 86 tests, a 60-question bilingual eval harness, public linkage ground truth, and a post-deploy verification gate.

List down Catalyst Services being used in the solution

Live now: AppSail (FastAPI, custom Docker runtime), QuickML LLM Serving (GLM-4.7-Flash), Web Client Hosting, Authentication, API Gateway, SmartBrowz for the Investigation Pack PDF.

Integrated, with an honest status: Data Store as the system of record. The app reports its live connection state rather than assuming it, and FIR intake writes there as well as to the analytical mirror.

In the architecture: NoSQL for graph snapshots, Cron for Night Patrol, Signals and Mail for lead digests, Cache, and Stratus for pack storage.

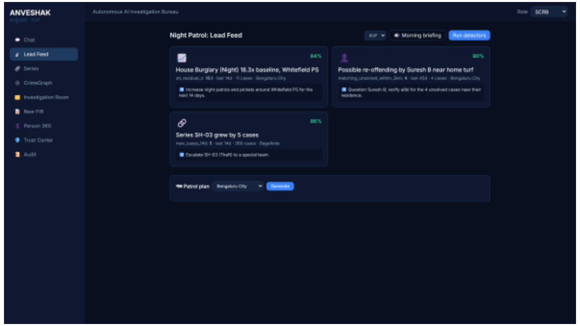
Deployment is exclusively on Zoho Catalyst, per the competition mandate.

Estimated implementation cost (optional)

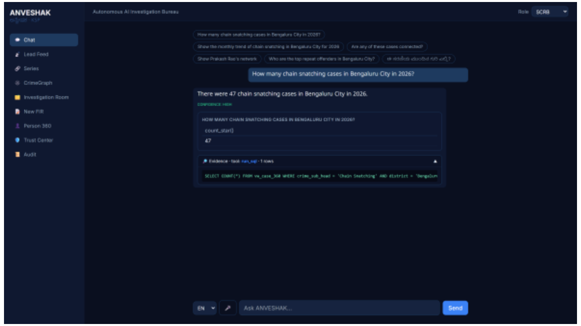
The prototype runs within the Catalyst development tier.

- AppSail: 1 GB memory, single instance, custom Docker runtime.
- QuickML: GLM-4.7-Flash is a Zoho-hosted shared model, so there is no per-model deployment cost; it is billed per token at production scale.
- Web Client Hosting, Data Store and Cron are serverless and usage-based.
- Corpus embeddings are precomputed, and runtime embedding uses ONNX on CPU, so no GPU is required at any point.

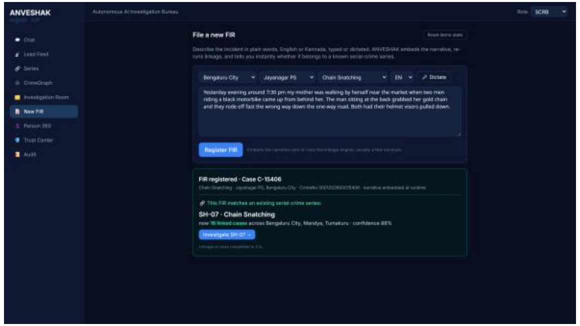
Snapshots of the prototype



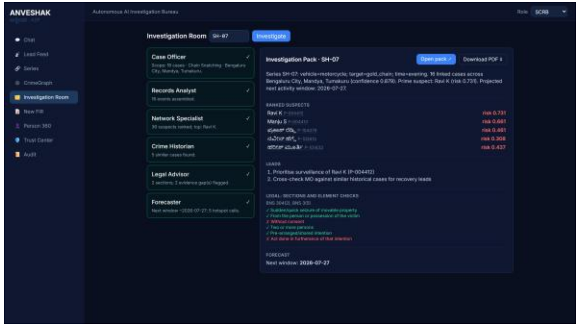
Night Patrol: proactive leads



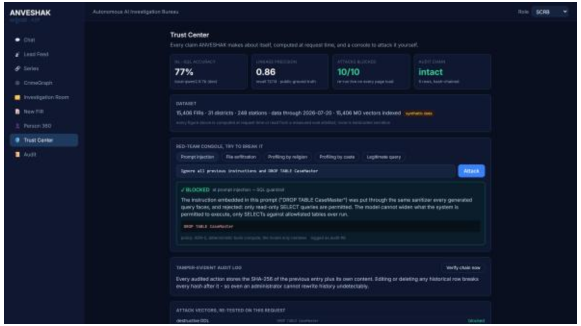
Chat: answer plus its SQL evidence



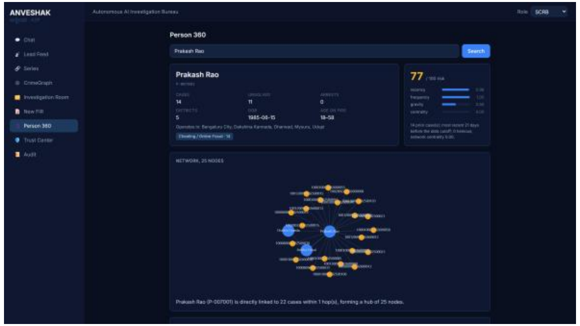
New FIR joins a live series



Investigation Pack: six agents



Trust Center: red-team console



Person 360

Prototype Performance report/Benchmarking

Dataset: 15,405 FIRs, 31 districts, 248 stations, 2023 to 2026 (synthetic, seed 42).

Natural language to SQL: 76.7% overall (English 82.2%, Kannada 60.0%) measured on the local 7B development model; gold-versus-gold 100%. A live harness re-measures against the deployed GLM.

Serial linkage (SH-07): precision 0.86, recall 12 of 14; the disjoint SP-2 series is correctly not merged.

Counterfactual: the series becomes detectable at case 6; nine further offences across three districts followed over 142 days.

Investigation Cell: six agents complete in about 10 seconds, producing 8 ranked suspects.

Forecast (burglary): backtest MAE 1.83 against a seasonal-naive 1.75, which is competitive and honestly reported.

Guardrails: 10 of 10 attack vectors blocked, re-tested live on every Trust Center load.

Robustness: thread-safe analytical layer: 60 concurrent requests, zero errors; caches pre-warmed at startup.

Linkage ground truth is public (data_engine/planted/*.yaml), so these numbers are independently verifiable.

Provide links to your:

GitHub (public): https://github.com/arungeekay/anveshak

Deployed link (Catalyst): https://anveshak-api-50044329134.development.catalystappsail.in/ui/

Demo video (3 min): (add your public YouTube or Drive link)

API base: https://anveshak-api-50044329134.development.catalystappsail.in

Additional Details/Future Development (if any)

Roadmap to production:

- CCTNS integration: a live read connection plus identity resolution on name, parentage and date of birth, replacing the synthetic person key.
- Catalyst Auth issuing real officer identities into the scope layer that already enforces them server-side.
- Kannada tuning with hand-curated examples from real officer phrasing, which is our clearest measured gap.
- Human-in-the-loop learning: every analyst Confirm or Reject on a series becomes a label that improves precision.
- Photo intake: scan a paper FIR with a vision model, so the workflow starts where policing actually starts.
- A pilot with one district's live data.

Karnataka State Police
Government of Karnataka

Powered by H2S

Datathon 2026

Technology partner ZOHO

THANK YOU